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Author manuscript

Ecotoxicol Environ Saf. Author manuscript; available in PMC 2023 July 01.

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Published in final edited form as:

Ecotoxicol Environ Saf. 2022 July 01; 239: 113604. doi:10.1016/j.ecoenv.2022.113604.

Circulating microRNAs as putative mediators in the association between short-term exposure to ambient air pollution and cardiovascular biomarkers

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Abstract

Background: Exposure to ambient air pollution is associated with increased cardiovascular morbidity and mortality. Circulating microRNAs (miRNAs) may mediate cardiovascular effects of exposure to air pollution. This study aims to investigate whether circulating miRNAs mediate the associations between short-term human exposure to ambient air pollution and cardiovascular biomarkers.

Methods: Twenty-four healthy adults residing in the Research Triangle area of North Carolina, USA were enrolled between December 2016 and July 2019. Circulating miRNAs, protein, and lipid biomarkers were assessed repeatedly for 3 sessions separated by at least 7 days. Linear mixed-effects models were used to assess the associations between air pollutant concentrations

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Disclaimer

The research described in this article has been reviewed by the Center for Public Health and Environmental Assessment, EPA, and approved for publication. The contents of this article should not be construed to represent agency policy nor does mention of trade names or commercial products constitute endorsement or recommendation for use.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2022.113604.

obtained from nearby air quality monitoring stations and miRNAs controlling for covariates including omega-3 index, relative humidity, and temperature. miRNAs that were significantly altered were then matched with protein or blood lipid biomarkers using either Ingenuity Pathway Analysis or a literature search. A mediation analysis was performed to test the statistical significance of miRNA's mediating effects between exposure to air pollution and cardiovascular biomarkers.

Results: Short-term exposure to ambient fine particulate matter (PM_{2.5}), ozone (O₃), and nitrogen dioxide (NO₂) was associated with changes in 11, 9, and 24 circulating miRNAs, respectively. Pathway analysis showed that several miRNAs including *miR-125b-5p*, *miR-144-5p*, *miR-26a-5p*, and *miR-34a-5p* may mediate the effects of air pollutant exposure on the changes of downstream protein / lipid biomarkers including serum amyloid A (SAA), C-reactive protein (CRP), soluble vascular adhesive molecules 1 (sICAM1), total cholesterol, and high-density lipoproteins (HDL). Mediation analysis showed that only *miR-26a-5p* significantly mediated air pollutant (PM_{2.5} and NO₂)-induced effects on blood CRP and total cholesterol levels. For example, 34.1% of PM_{2.5}-associated changes in CRP were significantly mediated by *miR-26a-5p* at lag4 [indirect effects, 0.06 (0.02, 0.10), *P* = 0.005]. Similarly, the proportions of indirect effects of *miR-26a-5p* on the association between NO₂ exposure and CRP were 46.8% at lag2 [0.06 (0.02, 0.11), *P* = 0.003], 61.2% at lag3 [0.05 (0.00, 0.09), *P* = 0.04], and 30.8% at 5-day moving average [0.06 (0.02, 0.10), *P* = 0.01]. In addition, omega-3 index may be a significant modifying factor of the mediated effects of miRNAs.

Conclusions: This study demonstrates that short-term exposure to ambient PM_{2.5}, O₃, and NO₂ was associated with specific circulating miRNAs, and some of which may mediate their effects on the downstream inflammation and blood lipid markers.

Keywords

microRNA; Mediation analysis; PM_{2.5}; O₃; NO₂; Cardiovascular; Inflammation; Blood lipids; Omega-3 polyunsaturated fatty acids

1. Introduction

Cardiovascular disease (CVD) is a leading cause of death worldwide (WHO, 2020). A growing body of evidence has shown that exposure to ambient air pollution is associated with increased cardiovascular morbidity and mortality. Short- and long-term exposure to air pollution is associated with elevated risk for myocardial infarction, stroke, heart failure, and arrhythmia (Brook et al., 2010, 2017; Cesaroni et al., 2014; Zhao et al., 2020). Proposed mechanisms of air pollution-induced cardiovascular effects include endothelial damage, oxidative stress, systemic inflammation, and autonomic nervous system imbalance (Chen et al., 2021a; Newby et al., 2015). It is currently believed that through these pathways, pulmonary exposure to air pollution may induce cardiovascular pathologies such as arteriosclerosis and acute coronary syndromes (Demetz and Ott, 2012; Foley and Conway, 2016).

These mechanisms may be regulated, in part, by the epigenetic mechanisms through microRNAs (miRNAs) (Alfano et al., 2018). miRNAs are a family of short (approximately

22 nucleotides), single-stranded, noncoding RNAs that are essential in post-transcriptional regulation of gene expression in cells (Macfarlane and Murphy, 2010). After biogenesis in original cells, miRNAs are secreted into circulating body fluids, such as extracellular vesicles (EV) in a stable and cell-free form in extracellular environments (Zhao et al., 2019). EV miRNAs in the circulation can be transported to recipient cells, relaying biological signals by regulating expression of target genes (Zhao et al., 2019). Thus, cell-to-cell communication realized by extracellular miRNAs is a possible mechanism through which respiratory exposure to air pollutants causes adverse effects in heart and blood vessels (Kao and Papoutsakis, 2019).

As recently described, exposure to ambient air pollutants, including fine particulate matter (PM_{2.5}), coarse particulate matter (PM₁₀), ozone (O₃), and nitrogen oxides (NO_x), was significantly associated with altered expression of circulating miRNAs, including *miR-146a-5p*, *miR-150-5p*, *miR-155-5p*, *miR-21-5p*, and *miR-25-3p* (H. Chen et al., 2020; Krauskopf et al., 2018; Motta et al., 2016; Rodosthenous et al., 2016; Ruiz-Vera et al., 2019). These miRNAs are predicted to target messenger RNAs (mRNA) of genes linked to cardiovascular pathology, such as interleukin 1 (*IL-1*), tumor necrosis factor (*TNFα*), C-reactive protein (*CRP*), and endothelin 1 (*ET-1*) (Chen et al., 2018; Rodosthenous et al., 2018). Mediation analysis showed that EV miRNAs, including *let-7c-5p*, *miR-331-3p*, *miR-185-5p*, *miR-106-5p*, and *miR-652-3p*, may mediate the association between PM₁₀ exposure and elevated blood fibrinogen levels (Pergoli et al., 2017). Although blood pressure is not a direct target of miRNA regulation, *miR-101* was a statistically significant mediator between PM₁₀ exposure and diastolic blood pressure changes among obese individuals (Motta et al., 2016). However, most of these analyses focused on single air pollutant or a limited number of cardiovascular biomarkers, there is no comprehensive investigation of the mediation role of miRNAs in the associations between exposure to multiple air pollutants (PM_{2.5}, O₃, and NO₂) and changes in a panel of cardiovascular biomarkers.

Previously, we reported that dietary omega-3 polyunsaturated fatty acids (n-3 PUFA) intake was associated with mitigation of adverse effects in respiratory and cardiovascular biomarkers linked to short-term exposure to low-level ambient air pollution in healthy adults (Chen et al., 2021b, 2022; Tong et al., 2021). In the present study, we hypothesize that exposure to ambient air pollutants, PM_{2.5}, O₃, and nitrogen dioxide (NO₂), may significantly alter expression of circulating miRNAs; and that the miRNAs whose expression is significantly altered mediate the association between exposure to air pollutants and cardiovascular biomarkers. Furthermore, we postulate that dietary n-3 PUFA intake modulates associations of those significantly identified mediational pathways.

2. Materials and methods

2.1. Study design

The study design has been previously described (Chen et al., 2021b; Tong et al., 2021). In brief, healthy participants residing in the Research Triangle area of North Carolina were recruited to this study from December 2016 to July 2019. Eligible participants met the following criteria: 25–55 years old; body mass index (BMI) between 19 and 35; having no history of cardiovascular disease, chronic respiratory disease, cancer,

uncontrolled hypertension (≥ 140 systolic, ≥ 90 diastolic), or diabetes; non-smokers for at least 1 year; not taking β -adrenergic receptor blockers, statins or anti-inflammatory drugs. Eligible participants were further screened and those meeting at least one of the following criteria were enrolled: 1) an in-house open-ended dietary questionnaire was used to screen participants whose eicosapentaenoic acid (EPA) + docosahexaenoic acid (DHA) intake was less than 0.5 g/week (low) or at least 3 g/week (high) for past 6 months or longer (Shen et al., 2019); 2) Omega-3 index (OmegaQuant, Sioux Falls, SD), a measurement of EPA and DHA in erythrocyte membrane, was employed to screen participants whose omega-3 index was less than 4.0% (low) or greater than 5.5% (high). Written informed consent was given by all participants prior to enrollment. The study was approved by the Institutional Review Board of the University of North Carolina at Chapel Hill and the U.S. Environmental Protection Agency and registered at [ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study?term=NCT02921048) (NCT02921048).

Each participant visited the U.S. Environmental Protection Agency's Human Studies Facility (HSF) for 2 consecutive days for up to 5 sessions with at least 1-week interval between two sessions. A total of 62 subjects (low omega-3 group: 28; high omega-3 group: 34) completed at least 3 study sessions. In order to conduct a cost-efficient preliminary assessment of the mediation role of miRNAs and possible moderation effects of n-3 PUFA, a subgroup of 12 participants (female: male = 1:1) with the lowest omega-3 indices (range: 3.18%–3.93%) in the low group and 12 (female: male = 1:1) with the highest omega-3 indices (5.5%–8.81%) in the high group were included in the analysis. The venous blood samples of the first 3 sessions of the 24 participants were analyzed in this study.

2.2. Exposure assessment

Hourly concentrations of ambient O_3 , $PM_{2.5}$, and NO_2 were obtained from central air monitoring stations (primary: Milbrook; alternative for missing data: Durham Armory) close to the HSF. Daily 24-hr average of $PM_{2.5}$ and NO_2 data were calculated from hourly pollutant data averaged between 9 AM and 8 AM the next day. Daily maximum 8-hr O_3 concentrations were calculated as the highest 8-h moving average concentrations between 9 AM to 8 AM the next day. Concentrations were obtained for each visit session, with the day of blood collection as lag0, 1–4 days prior as lag1–lag4, and the 5-day moving average (5dMA). Daily 24-hr average of relative humidity (RH) and temperature were also collected from the same monitoring station.

2.3. Biomarker measurements

2.3.1. Measurement of blood lipids, endothelin-1 and markers of fibrinolysis and inflammation—

Blood lipids were measured by LabCorp (Burlington, NC). Commercially available multiarray kits were used to quantify plasma levels of interleukin 1 β (IL-1 β), IL-6, IL-8, tumor necrosis factor alpha (TNF α), C-reactive protein (CRP), soluble vascular cell adhesion molecule 1 (sVCAM1), soluble intercellular adhesion molecule 1 (sICAM1), serum amyloid A (SAA), tissue-type plasminogen activator (tPA), plasminogen, D-dimer, and von Willebrand factor (vWF) (Meso-Scale Diagnostics, Gaithersburg, MD). In addition, vasoconstrictor endothelin-1 (ET-1) and oxidized low-density lipoprotein (Ox-LDL) were measured using kits purchased from Bachem (Torrence,

CA) and Mercodia (Winston Salem, NC), respectively. All assays were carried out according to the manufacturer's instructions.

2.3.2. miRNA profiling—The multiplex miRNA assay was carried out using FirePlex® particle technology (ABCAM, Cambridge, MA). The FirePlex® cardiology panel profiles 65 miRNAs differentially regulated in normal or abnormal cardiovascular health. Briefly, 40 µL plasma sample was prepared by mixing with Digest Buffer and Protease Mix to a final volume of 80 µL and then incubated at 60 °C for 45 min. In a 96-well filter plate, 25 µL of the prepared sample was incubated with 35 µL FirePlex Particles and 25 µL of hybridization buffer at 37 °C for 60 min. After rinsing to remove unbound RNA, 75 µL of labeling buffer was added to each well, and the plate was incubated for 60 min at room temperature. Adapted-modified miRNAs were eluted using 95 °C water and collected for PCR amplification using a PCR master mix. PCR product was then hybridized with hybridization buffer at 37 °C for 30 min. Particles were then rinsed and scanned using an EMD Millipore Guava 8HT flow cytometer. Raw output was background subtracted, normalized using the geometric mean of the three normalizer miRNAs.

2.4. Data analysis

The data analysis was performed using SAS 9.4 software (SAS Institute Inc, Cary, NC). To improve normality in the residuals, we log-transformed (log2) miRNA and biomarker data for the association analysis. Generalized linear mixed model with random subject intercepts were employed for the statistical modeling and the models were adjusted for omega-3 index, age, sex, race, body mass index (BMI), long-term trend (natural spline of date), day of week, relative humidity, and temperature (Chen et al., 2021b; Krauskopf et al., 2019; Rodosthenous et al., 2018). PM_{2.5}, O₃ and NO₂ were considered as an immediate (lag0), delayed (lag1 to lag4), or cumulative (5dMA) linear effect. The steps of the statistical analysis are described in a flowchart (Fig. 1A) and detailed below.

1. The first step was to identify miRNAs that were significantly associated with short-term exposure to ambient PM_{2.5}, O₃ or NO₂. We assumed that only air pollutant associated miRNAs could further mediate the effects on the downstream cardiovascular protein or lipid markers.
2. Secondly, an *in-silico* analysis was conducted to find validated mRNA targets of the miRNAs identified in Step 1 using the Ingenuity Pathway Analysis (Ingenuity Systems®, Redwood City, CA). The miRNA–mRNA matches were only kept for those protein biomarkers that have been measured in this study. We then performed a literature search to identify links between miRNAs and blood lipid targets including total cholesterol, high density lipoprotein (HDL), low density lipoprotein (LDL), D-dimer, or ox-LDL. After this step, we established theoretical mediation pathways, i.e., air pollutant – miRNA – protein / lipid biomarkers.
3. In the third step, the established pathways in Step 2 were further assessed using the experimental data to validate whether the mediation effects of miRNAs are statistically significant. To assess the mediation of miRNAs, we estimated the statistical significance of the indirect effects of miRNAs on the association

between exposure to air pollutants and biomarkers. This approach decomposes the total estimated effects of air pollutants on the biomarker expression into an estimate of the direct effect and an estimate of the indirect effect of air pollutants on the biomarker expression mediated by miRNA. To allow for a comparable analysis, we standardized the data of exposure (air pollutants), mediator (miRNAs), and outcomes (protein and lipid biomarkers) by subtracting the mean and dividing by the standard deviation. Two models have been fitted using the generalized linear mixed model as described above. As shown in Fig. 1B, the first model ($X \rightarrow M$) is to assess the association between exposure to air pollutants (X) and miRNAs (M). The second model ($X + M \rightarrow Y$) is to assess the link between miRNAs and protein / lipid biomarkers (Y) while also considering air pollutant (X). Sobel test was conducted to evaluate if the indirect effect was statistically significant (Sobel, 1986). The results of $P < 0.05$ and < 0.10 are considered as “significant” and “borderline significant”, respectively.

When interpreting mediation results, it is possible that the change directions were different between direct and indirect effects. This phenomenon was termed as suppression where the statistical removal of a mediational effect increased the magnitude of the relationship between independent and dependent variables (MacKinnon et al., 2000). In this study, we only focused on those with consistent mediational results, namely the same directions between direct and indirect effects.

4. In the fourth step, we further evaluated whether omega-3 index modified the significant mediation pathways that were discovered in Step 3. As shown in Fig. 1B, we added an interaction of exposure and modifier (omega-3 index, W) to the model between air pollutants (X) and miRNAs (M) ($X + W + X * W \rightarrow M$). Similarly, the interaction ($X * W$) was also added into model between air pollutants (X), miRNAs (M) and protein / lipid biomarkers (Y) ($X + W + X * W + M \rightarrow Y$). Here we report the p-values of the interaction. If $P < 0.05$, it indicates that increases in omega-3 index significantly changed the association between exposure to air pollutant and miRNA levels or the association between exposure to air pollution and biomarkers.

3. Results

3.1. Descriptive statistics

Supplemental Table 1 shows salient characteristics of the 24 participants in this study. The participants were 39 years old in average and half of them were male. Majority of the participants were caucasians (75%). All participants were non-smokers and were not taking β -adrenergic receptor blockers, anti-inflammatory drugs, or statins. During the study period, the daily maximum 8-hr O_3 was 41.8 ppb with a standard deviation (SD) of 11.2 ppb. Average daily level of $PM_{2.5}$ was $9.8 \pm 3.6 \mu g/m^3$, while the NO_2 concentration was 4.8 ± 3.4 ppb (Supplemental Table 2). These air pollutant levels were well below the National Ambient Air Quality Standards (NAAQS). The correlations between air pollutant concentrations and meteorological measurements (temperature and relative humidity) ranged

between weak and moderate (Supplemental Table 2). Descriptive statistics of all examined miRNAs, protein markers and blood lipids were summarized in Supplemental Table 3.

3.2. Association between exposure to air pollutants and expression of circulating miRNAs

As a first approach, we assumed that only miRNAs that were significantly associated with air pollutant exposure may mediate air pollutant induced biological effects. Among the 65 measured miRNAs, short-term exposure to ambient PM_{2.5}, O₃, and NO₂ was associated with expression changes in 11, 9, and 24 miRNAs, respectively (Supplemental Table 4–6). For consistency across all modeled results, we only report coefficients for the associations and presented examples of the significant associations for those miRNAs that were predicted to target protein and / or lipid markers measured here in the main text. For example, PM_{2.5} was associated with elevated expression in *miR-124-3p* at lag4 [coefficient and 95% confidence interval (95% CI): 0.16 (95% CI: 0.04, 0.28)], *miR-125b-5p* at lag4 [0.18 (95% CI: 0.03, 0.33)] and 5dMA [0.18 (95% CI: 0.01, 0.35)], *miR-194-5p* at lag4 [0.07 (95% CI: 0.002, 0.13)], *miR-26a-5p* at lag4 [0.04 (95% CI: 0.01, 0.06)], and *miR-34a-5p* at lag4 [0.15 (95% CI: 0.03, 0.28)], as well as decreased expression in *miR-320b* at lag1 [– 0.02 (95% CI: –0.04, –0.004)] (Fig. 2). Short-term exposure to O₃ was associated with increased expression in *let-7e-5p* at lag2 [0.04 (95% CI: 0.02, 0.07)] and 5dMA [0.05 (95% CI: 0.01, 0.09)], *miR-125b-5p* at lag1 [0.07 (95% CI: 0.01, 0.14)] and 5dMA [0.15 (95% CI: 0.06, 0.24)], and *miR-26a-5p* at lag2 [0.01 (95% CI: 0.0004, 0.02)] (Fig. 3). In addition, ambient NO₂ was significantly associated with elevated expression in *let-7e-5p* at lag3 [0.08 (95% CI: 0.03, 0.13)] and 5dMA [0.13 (95% CI: 0.03, 0.24)], *miR-146a-5p* at lag4 [0.03 (95% CI: 0.01, 0.05)] and 5dMA [0.07 (95% CI: 0.03, 0.11)], *miR-199a-5p* at 5dMA [0.05 (95% CI: 0.002, 0.1)], *miR-26a-5p* at lag2 [0.05 (95% CI: 0.02, 0.08)], lag3 [0.03 (95% CI: 0.003, 0.05)], lag4 [0.04 (95% CI: 0.02, 0.06)], and 5dMA [0.09 (95% CI: 0.05, 0.13)], and *miR-328-3p* lag4 [0.05 (95% CI: 0.005, 0.1)] and 5dMA [0.12 (95% CI: 0.03, 0.22)]. Short-term exposure to NO₂ was also associated with decreases in *miR-144-5p* expression at lag3 [– 0.19 (95% CI: –0.32, –0.06)] and 5dMA [–0.33 (95% CI: –0.63, –0.02)] (Fig. 4).

3.3. Predict and match downstream protein and lipid targets of miRNAs

In this study, we measured blood lipids and plasma protein levels of inflammation, coagulation, and endothelial function markers (i.e. IL-1 β , IL-6, IL-8, TNF α , CRP, SAA, vWF, tPA, D-dimer, sICAM1, sVCAM1, and ET-1). As shown in the supplemental Table 7, there were some associations between air pollutant and protein / lipid biomarkers with current sample size. However, the mediational effects of the mediator (i.e. miRNA) could still occur even if the independent variable (i.e., air pollutant) was not significantly associated with the dependent variable (i.e., protein / lipid biomarkers) (MacKinnon et al., 2002). Thus, mediational analyses were conducted on all identified air pollutant – miRNA – protein / lipid biomarker matches.

The second step was to identify downstream protein or lipid targets of air pollutant-associated miRNAs. To explore the possible links between air pollutant associated miRNAs and these downstream protein targets, we first conducted a pathway analysis using the Ingenuity Pathway Analysis for protein and literature search for lipid biomarkers (Moore

et al., 2011; Qin and Guo, 2020; Rotllan et al., 2016; Sun et al., 2012). As shown in Table 1, of the PM_{2.5}-associated miRNAs, seven (i.e. *miR-124-3p*, *miR-125b-5p*, *miR-194-5p*, *miR-26a-5p*, *miR-320b*, *miR-34a-5p*, and *miR-370-3p*) were projected to regulate the changes of IL-8, TNF α , SAA, ET-1, CRP, D-dimer, sVCAM1, total cholesterol, and HDL. Three O₃-associated miRNAs (i.e. *let-7e-5p*, *miR-125b-5p*, and *miR-26a-5p*) may regulate the changes in IL-6, IL-8, CRP, ET-1, SAA, total cholesterol, and HDL. In addition, nine NO₂-linked miRNAs (i.e. *let-7e-5p*, *miR-106b-5p*, *miR-124-3p*, *miR-126-3p*, *miR-144-5p*, *miR-146a-5p*, *miR-199a-5p*, *miR-26a-5p*, and *miR-328-3p*) could modify the downstream in IL-1 β , IL-6, IL-8, TNF α , CRP, ET-1, SAA, sVCAM1, total cholesterol, Ox-LDL, and HDL.

3.4. Mediation effects of miRNAs on the association between exposure to air pollutants and biomarkers

Once mediation pathways of air pollutant – miRNA – protein / lipid biomarkers were established, we conducted a mediation analysis to evaluate whether the mediation effects of miRNAs were statistically significant. The data of exposure (X, air pollutants), mediator (M, miRNAs), and outcomes (Y, protein / lipid biomarkers) were normalized by subtracting the mean and dividing by the standard deviation for comparable results before the mediation analysis. The mediation statistics for all identified pathways for air pollutant – miRNA – biomarkers are presented in Supplemental Table 8. There were 8, 4 and 12 mediation results with at least borderline significance ($P < 0.1$) for PM_{2.5}, O₃, and NO₂, respectively (Table 2). A further examination indicated that, among the 24 “statistically significant” mediation effects, nine results showed inconsistent directions between direct and indirect effects, constituting possible suppression. Here, we only reported the coefficients of the indirect effects [estimates (95% CI)] with significant mediation statistics ($P < 0.05$) that showed the same directional changes as direct effects. As shown in Table 2, expression changes in circulating *miR-26a-5p* was significantly mediating approximately 34.1% of PM_{2.5} exposure-induced effects on CRP at lag4 [0.06, (0.02, 0.10)]. Similarly, circulating *miR-26a-5p* may significantly mediate 46.8%, 61.2%, and 30.8% NO₂-induced changes in CRP at lag2 [0.06 (0.02, 0.11)], lag3 [0.05 (0.00, 0.09)], and 5dMA [0.06 (0.02, 0.10)]. Changes in circulating *miR-26a-5p* may also significantly mediate 68.6% and 46.5% of the effects of NO₂ exposure on blood total cholesterol levels at lag3 [– 0.04 (–0.07, –0.003)] and lag4 [– 0.04 (–0.06, –0.01)].

3.5. Moderation effects of omega-3 index on the association between air pollutants and miRNAs or downstream biomarkers

Participants in this study were categorized according to their circulating levels of omega-3 fatty acids. We further investigated the modifying effects of omega-3 PUFA on the air pollutant induced changes in biomarkers by introducing an interaction term between air pollutant and omega-3 index on the identified significant mediation pathways between air pollutants and downstream protein and lipid markers. As summarized in Fig. 5, omega-3 index did not significantly modify PM_{2.5} induced changes in *miR-26a-5p* at lag4 [0.00 (–0.06, 0.05)] but significantly affected PM_{2.5} induced changes in plasma CRP concentrations at lag4 [0.11 (0.08, 0.14)]. The associations between NO₂ exposure and changes in circulating *miR-26a-5p* were significantly modified by omega-3 index at lag2 [–

0.22 (−0.3, −0.14)], lag3 [− 0.30 (−0.38, −0.22)], and lag4 [0.07 (0.01, 0.12)]. Similarly, the associations between NO₂ exposure and changes in plasma CRP concentrations were significantly moderated by omega-3 index at lag3 [0.18 (0.13, 0.23)] and 5dMA [0.08 (0.03, 0.12)]. Finally, omega-3 index also significantly modified the association between NO₂ and blood total cholesterol levels at lag3 [− 0.03 (−0.06, −0.003)] and lag4 [− 0.04 (−0.05, −0.02)] (Fig. 5).

4. Discussion

Exposure to ambient air pollution has been associated with elevated risk of adverse cardiovascular effects. Circulating miRNAs may play an important role in bridging pulmonary exposure to air pollution and cardiovascular effects. In the current study, we explored whether exposure to ambient air pollution induce changes in cardiovascular biomarkers of inflammation, coagulation, and blood lipids through alterations in circulating miRNAs among healthy adults. We report that *miR-26a-5p* significantly relay the effects of short-term exposure to ambient air pollutants to downstream plasma CRP and blood total cholesterol levels.

One of the proposed mechanisms underlying the cardiovascular effects of air pollution exposure was through extracellular miRNAs that could act as messengers to relay biological signals from exposed cells in the airway to the target cells in the cardiovascular system (Newby et al., 2015; Zhao et al., 2019). miRNAs are known to regulate mRNA translation through a biological complex known as RNA-inducing silencing complex (RISC). This complex can regulate target mRNA translation through a post-initiation step without reducing mRNA abundance, polyadenylation, or polyribosomal content, a cap-dependent and poly (A)-dependent mechanism, or inducing de-adenylation of mRNAs independently of inhibition of translation (Abdellatif, 2012). Changes in miRNA expression had been linked to the pathophysiology of cardiovascular diseases, ranging from cardiovascular risk factors (e.g., hypertension, diabetes, dyslipidemia), to atherosclerosis, myocardial infarction, and cardiac remodeling (Peters et al., 2020). For example, *miR-126* could inhibit chemokine CXCL12-induced angiogenesis through SPRED-1 stimulation in human umbilical vein endothelial cells (Bassand et al., 2021). *miR-146* may downregulates small ubiquitin-like modifier 1 expression, which was essential in restoring SUMOylation and cardiac function (Oh et al., 2018). Thus, it is possible that circulating miRNAs were a significant factor bridging exposure to air pollutants to adverse cardiovascular outcomes through post-transcriptional regulation of miRNAs.

Focusing on a pathway – centered approach, we employed a panel of 65 miRNAs that have been shown to be linked to cardiovascular diseases. We observed that short-term exposure to PM_{2.5}, O₃, and NO₂ was significantly associated with changes in multiple circulating miRNAs including *miR-126-3p*, *miR-146a-5p*, *miR-199a-3p*, *miR-25-3p*, and *miR-26a-5p*. Our results were consistent with the literature reporting an association between exposure to ambient air pollution and expression changes in circulating miRNAs, which has been well documented (Farraj et al., 2011; Krauskopf et al., 2018; Rodosthenous et al., 2016; Ruiz-Vera et al., 2019). A study demonstrated that long-term exposure to ambient PM_{2.5} among 22 men aged 60 years and older was associated with altered expression of

serum miRNAs including *miR-126-3p*, *miR-146a-5p*, *miR-19b-3p*, *miR-191-5p*, *miR-150-5p*, and *miR-15a-5p*, implicating cardiovascular disease related pathways such as oxidative stress, inflammation, and atherosclerosis (Rodosthenous et al., 2016). Short-term exposure to ambient O₃ was related to significant changes in EV *miR-150* and *miR-155* among coronary artery disease patients (H. Chen et al., 2020). In a controlled exposure study, acute exposure of healthy volunteers to 400 ppb O₃ for 2 h induced altered expression of several miRNAs (e.g. *miR-143*, *miR-145*, *miR-199a*, *miR-222*, and *miR-25*) in the sputum samples, associating with pro-inflammatory responses in the lung (Fry et al., 2014). In a crossover study, short-term exposure to traffic-related air pollution, including NO₂, from a busy street was associated with altered expression in miRNAs, including *miR-145-5p*, *miR-146-5p*, and *miR-25-3p* (Krauskopf et al., 2018, 2019).

We then matched significantly associated miRNAs with the downstream protein / lipid targets that were measured in the study. There were 13, 8, and 20 “air pollutant – miRNA – protein / lipid biomarker” pathways identified for PM_{2.5}, O₃, and NO₂, respectively. These identified pathways implicate multiple mechanisms involved in cardiovascular pathophysiology, including systemic inflammation, vascular inflammation and injury, coagulopathy, and dyslipidemia. (Bassand et al., 2021). However, these projected pathways warrant further statistical validation using the mediation analysis. Our mediation results suggested that only circulating *miR-26a-5p* may significantly relay the effects of exposure to air pollutants on the downstream cardiovascular biomarkers. Specifically, both PM_{2.5} and NO₂ associated increases in *miR-26a-5p* were mediating approximately 30–60% upregulation of CRP levels. CRP is an acute inflammatory protein that indicates early phase of immune responses of the host and a predictive biomarker for cardiovascular disease (Sproston and Ashworth, 2018). *miR-26a* is believed to be heavily involved in innate immune response signaling, including a reciprocal inhibition between *miR-26a* and NFκB to regulate inflammatory responses (Xie et al., 2015). These findings indicated elevated expression of *miR-26a-5p* was involved in the early acute immune responses, such as elevated CRP levels, in response to ambient air pollution exposure. On the other hand, NO₂ associated elevations in *miR-26a-5p* accounted for 45 – 70% of total cholesterol reduction, suggesting a possible protective mechanism of *miR-26a-5p* in lowering blood total cholesterol levels. Interestingly, miR-26a has been shown to downregulate blood cholesterol levels and atherosclerosis progression through inhibiting transient receptor potential canonical 3 (Feng et al., 2018). Our findings may direct future research to investigate *miR-26a*'s role in air pollution exposure induced cardiovascular effects through regulation of inflammation and lipid metabolism.

Although the indirect effects were statistically significant for multiple identified mediational pathways (e.g., PM_{2.5} lag4 – *miR-125b-5p* – SAA; PM_{2.5} lag4 – *miR-34a-5p* – sVCAM1; O₃ lag1 – *miR-125b-5p* – SAA; NO₂ lag3 – *miR-144-5p* – SAA; NO₂ lag4 – *miR-26a-5p* – CRP), we observed inconsistency in the directions of changes between direct and indirect effects, suggesting possible suppression. This pattern goes against the statistical assumption that the indirect effects of the mediator are only part of the total effects. The suppression effects of miRNA derived from the statistical model may bear biological significance and warrants further investigation. Nevertheless, only a few other studies have utilized a similar mediation analysis to examine miRNA's role in air pollution induced biological effects. For

instance, in a double-blind, randomized crossover trial with young healthy adults, Chen and colleagues reported that circulating miRNAs (i.e., *miR-1-3p*, *miR-187-3p*, *miR-146a-5p*) may mediate (borderline significant) the effects of PM_{2.5} on mRNA expression of serum *ET-1* and *TNFA* (Chen et al., 2018). In another observational study, serum levels of benzo [a]pyrene-r-7,t-8,t-9, c-10-tetrahydrotetrol-albumin (BPDE-Alb) adducts, a proxy of internal dose of PM exposure, induce elevated mRNA expression of *IL-6* and toll-like receptor 2 (*TLR2*) in children, possibly mediated by the regulation of *let-7a*, *miR-146a-5p*, and *miR-155-5p* (Li et al., 2020). Mechanistic studies, on the other hand, corroborated a possible role of miRNAs in air pollutant induced cardiovascular effects. For example, ammonia inhalation induced cardiac autophagy which may be mediated through regulation of the *miR-202-5p* and the PTEN/AKT/mTOR pathway (Xing et al., 2019). *miR-6615-5p* and *miR-15a* were both associated with to NFκB pathway to induce inflammatory injuries and oxidative stress mediated apoptosis of ammonia gas exposure (D. Chen et al., 2020; Han et al., 2020).

We also evaluated the modifying effects of dietary intake of omega-3 PUFA on the mediational pathways. Dietary intake of omega-3 PUFA, especially EPA and DHA, has respiratory and cardiovascular protection against adverse effects associated with air pollution exposure (Chen et al., 2021b; Lin et al., 2019; Tong et al., 2012, 2021). In this study, we found that higher dietary intake of omega-3 PUFA may significantly modify both direct and indirect effects between air pollutant exposure and cardiovascular biomarkers. Of particular interest, omega-3 index modified the association between NO₂ exposure and circulating *miR-26a-5p*, suggesting that omega-3 PUFA may affect the association between NO₂ exposure and protein / lipid biomarkers through possible moderated mediation of *miR-26a-5p*. In addition, omega-3 index positively affected the direct association between exposure to PM_{2.5} or NO₂ and acute phase proteins CRP at delayed lags (e.g., lag3 and lag4). This may be explained by the notion that extended exposure to oxidative pollutants may lead to oxidation of omega-3 PUFA, negating its antioxidant properties and leading to inflammation. In addition, omega-3 PUFA appeared to negatively impact the association between NO₂ exposure and total cholesterol levels, suggesting a possible blood lipid lowering effects of omega-3 PUFA.

This study is the first to explore the role of circulating miRNAs in regulating lower-than-NAAQS-level ambient air pollution induced cardiovascular effects among health adults. We measured miRNA, protein, and lipid biomarkers simultaneously in our study, allowing for a preliminary assessment of the potential mediation effects of miRNAs in air pollution induced cardiovascular effects. However, there are limitations to this study. First, the sample size of this study is relatively small with only 24 participants in 3 repeated measurements and the *P* values were not adjusted for multiple testing, increasing the chance of both type I and type II errors. However, this is an exploratory study showing a possible mediation role of miRNAs and pointing to a future research direction, thus increasing sensitivity of the statistical testing is justified. Second, we employed air monitoring station data for air pollutants rather than individual exposure metrics such as location, time spent indoor vs. outdoor, and activity level, which could possibly introduce non-differential exposure misclassification. Third, we cannot assume causal relationships of the air pollutant – miRNA – protein / lipid pathways based on the association results of the mediation analysis. Fourth,

other mechanisms of air pollution induced cardiovascular effects may not be accounted for in current analysis. Fifth, although we have controlled covariates such as temperature, relative humidity, omega-3 index, other factors, such as lifestyle (e.g., exercise, dietary factors) and exposure to other pollutants (e.g., secondary organic aerosols), are possible confounders.

5. Conclusions

This study evaluated the effects of short-term exposure to ambient air pollution below the NAAQS on circulating miRNAs and their possible mediating effects on cardiovascular biomarkers in healthy adults in North Carolina, USA. We have found that multiple circulating miRNAs were significantly associated with exposure to ambient PM_{2.5}, O₃, and NO₂. Among the significantly affected miRNAs, circulating *miR-26a-5p* may significantly mediate the effects of air pollutants on the downstream biomarkers such as CRP and total cholesterol. These findings may offer preliminary evidence that cardiovascular effects induced by short-term exposure to ambient air pollution may be mediated through circulating miRNAs.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgements

The U.S. Environmental Protection Agency Intramural Research Program supported this research.

Acknowledgement of other contributors

We thank Claudia Salazar, Tracey Montilla, Julie Wood, Martin Case, Lisa Dailey, Joleen Soukup, and Shirley Harder for their excellent technical support. We thank the recruitment service from MPF Federal. This project was supported in part by an appointment to the Research Participation Program at the U.S. Environmental Protection Agency, administered by the Oak Ridge Institute for Science and Education through an interagency agreement between the U.S. Department of Energy and EPA.

Abbreviations:

CRP	C-reactive protein
ET-1	endothelin 1
HDL	high-density lipoprotein
IL-1β	interleukin 1 beta
LDL	low-density lipoprotein
Ox-LDL	oxidized low-density lipoprotein
SAA	serum amyloid A
sICAM1	soluble intercellular adhesion molecule 1
sVCAM1	soluble vascular adhesion molecule 1

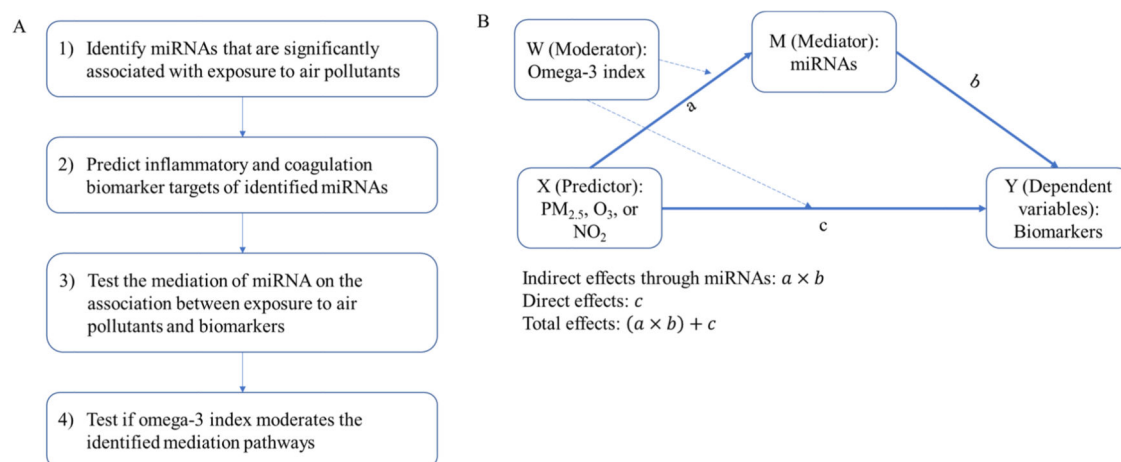
TNFα	tumor necrosis factor alpha
tPA	tissue plasminogen activator
vWF	von Willebrand Factor

References

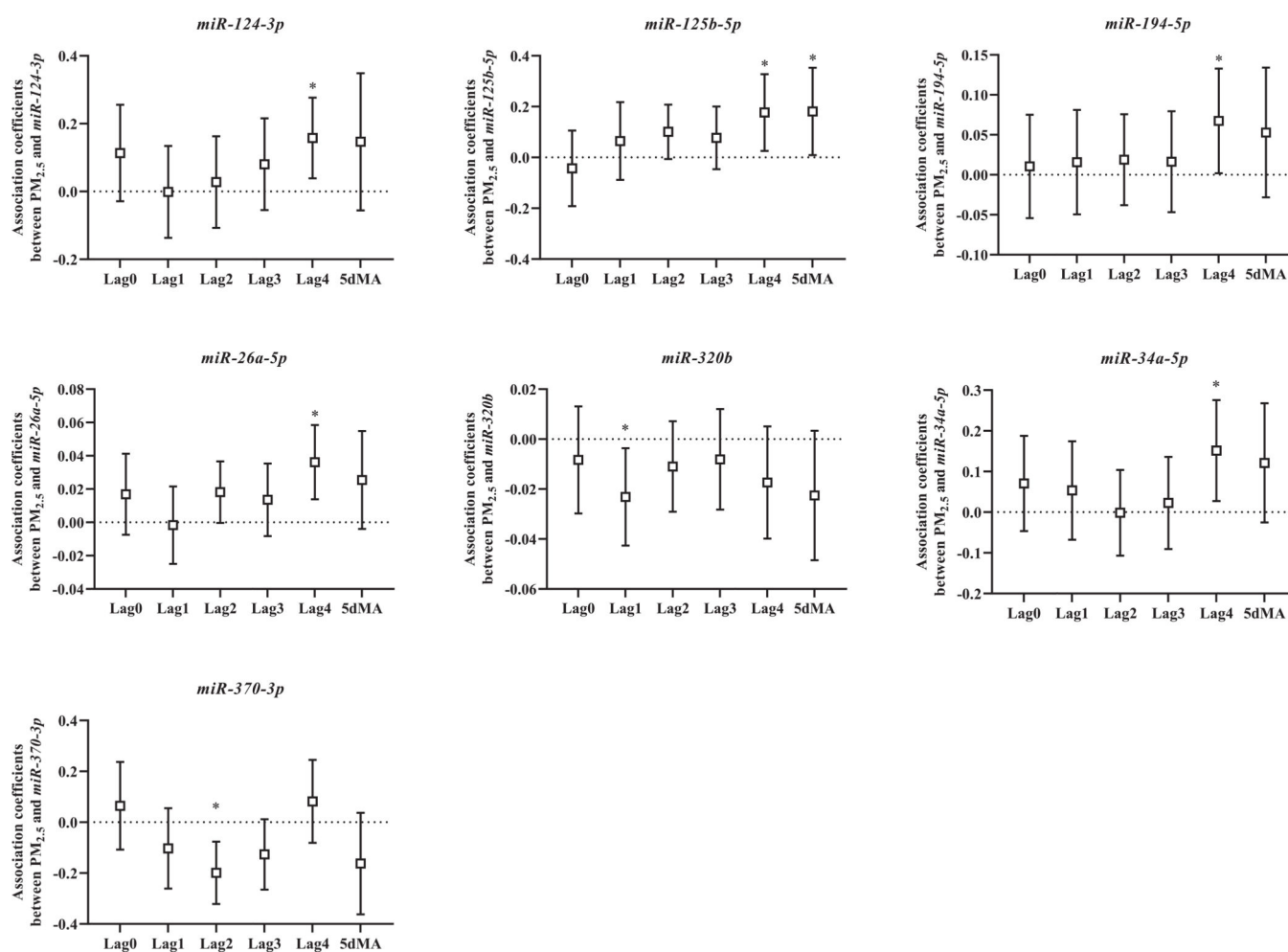
- Abdellatif M, 2012. Differential expression of micrnas in different disease states. *Circ. Res.* 110, 638–650. [PubMed: 22343558]
- Alfano R, Herceg Z, Nawrot TS, Chadeau-Hyam M, Ghantous A, Plusquin M, 2018. The impact of air pollution on our epigenome: how far is the evidence? (a systematic review). *Curr. Environ. Health Rep.* 5, 544–578. [PubMed: 30361985]
- Bassand K, Metzinger L, Naim M, Mouhoubi N, Haddad O, Assoun V, et al. , 2021. Mir-126–3p is essential for cxcl12-induced angiogenesis. *J. Cell Mol. Med.* 25 (13), 6032–6045.
- Brook RD, Rajagopalan S, Pope CA 3rd, Brook JR, Bhatnagar A, Diez-Roux AV, et al. , 2010. Particulate matter air pollution and cardiovascular disease: an update to the scientific statement from the american heart association. *Circulation* 2331–2378. [PubMed: 20458016]
- Brook RD, Newby DE, Rajagopalan S, 2017. The global threat of outdoor ambient air pollution to cardiovascular health: time for intervention. *JAMA Cardiol.* 2, 353–354. [PubMed: 28241232]
- Cesaroni G, Forastiere F, Stafoggia M, Andersen ZJ, Badaloni C, Beelen R, et al. , 2014. Long term exposure to ambient air pollution and incidence of acute coronary events: prospective cohort study and meta-analysis in 11 european cohorts from the escape project. *BMJ* 348, f7412. [PubMed: 24452269]
- Chen D, Ning F, Zhang J, Tang Y, Teng X, 2020. Nf-kappab pathway took part in the development of apoptosis mediated by mir-15a and oxidative stress via mitochondrial pathway in ammonia-treated chicken splenic lymphocytes. *Sci. Total Environ.* 729, 139017. [PubMed: 32380330]
- Chen H, Xu Y, Rappold A, Diaz-Sanchez D, Tong H, 2020. Effects of ambient ozone exposure on circulating extracellular vehicle microrna levels in coronary artery disease patients. *J. Toxicol. Environ. Health A* 1–12.
- Chen H, Samet JM, Bromberg PA, Tong H, 2021a. Cardiovascular health impacts of wildfire smoke exposure. *Part. Fibre Toxicol.* 18, 2. [PubMed: 33413506]
- Chen H, Zhang S, Shen W, Salazar C, Schneider A, Wyatt L, et al. , 2021b. The influence of dietary intake of omega-3 polyunsaturated fatty acids on the association between short-term exposure to ambient nitrogen dioxide and respiratory and cardiovascular outcomes among healthy adults. *Environ. Health* 20, 123. [PubMed: 34872587]
- Chen H, Zhang S, Shen W, Salazar C, Schneider A, Wyatt LH, et al. , 2022. Omega-3 fatty acids attenuate cardiovascular effects of short-term exposure to ambient air pollution. *Part. Fibre Toxicol.* 19, 12. [PubMed: 35139860]
- Chen R, Li H, Cai J, Wang C, Lin Z, Liu C, et al. , 2018. Fine particulate air pollution and the expression of micrnas and circulating cytokines relevant to inflammation, coagulation, and vasoconstriction. *Environ. Health Perspect.* 126, 017007. [PubMed: 29342453]
- Demetz G, Ott I, 2012. The interface between inflammation and coagulation in cardiovascular disease. *Int. J. Inflam.* 2012, 860301. [PubMed: 22518344]
- Farraj AK, Hazari MS, Haykal-Coates N, Lamb C, Winsett DW, Ge Y, et al. , 2011. St depression, arrhythmia, vagal dominance, and reduced cardiac micro-rna in particulate-exposed rats. *Am. J. Respir. Cell Mol. Biol.* 44, 185–196. [PubMed: 20378750]
- Feng M, Xu D, Wang L, 2018. Mir-26a inhibits atherosclerosis progression by targeting trpc3. *Cell Biosci.* 8, 4. [PubMed: 29387339]
- Foley JH, Conway EM, 2016. Cross talk pathways between coagulation and inflammation. *Circ. Res.* 118, 1392–1408. [PubMed: 27126649]
- Fry RC, Rager JE, Bauer R, Sebastian E, Peden DB, Jaspers I, et al. , 2014. Air toxics and epigenetic effects: ozone altered micrnas in the sputum of human subjects. *Am. J. Physiol. Lung Cell Mol. Physiol.* 306, L1129–L1137. [PubMed: 24771714]

- Han Q, Tong J, Sun Q, Teng X, Zhang H, Teng X, 2020. The involvement of mir-6615–5p/smad7 axis and immune imbalance in ammonia-caused inflammatory injury via nf-kappab pathway in broiler kidneys. *Poult. Sci.* 99, 5378–5388. [PubMed: 33142454]
- Kao CY, Papoutsakis ET, 2019. Extracellular vesicles: exosomes, microparticles, their parts, and their targets to enable their biomanufacturing and clinical applications. *Curr. Opin. Biotechnol.* 60, 89–98. [PubMed: 30851486]
- Krauskopf J, Caiment F, van Veldhoven K, Chadeau-Hyam M, Sinharay R, Chung KF, et al. , 2018. The human circulating mirnome reflects multiple organ disease risks in association with short-term exposure to traffic-related air pollution. *Environ. Int.* 113, 26–34. [PubMed: 29421404]
- Krauskopf J, van Veldhoven K, Chadeau-Hyam M, Vermeulen R, Carrasco-Turigas G, Nieuwenhuijsen M, et al. , 2019. Short-term exposure to traffic-related air pollution reveals a compound-specific circulating mirna profile indicating multiple disease risks. *Environ. Int.* 128, 193–200. [PubMed: 31059914]
- Li J, Wang T, Wang Y, Xu M, Zhang L, Li X, et al. , 2020. Particulate matter air pollution and the expression of micrnas and pro-inflammatory genes: association and mediation among children in Jinan, China. *J. Hazard. Mater.* 389, 121843. [PubMed: 31843406]
- Lin Z, Chen R, Jiang Y, Xia Y, Niu Y, Wang C, et al. , 2019. Cardiovascular benefits of fish-oil supplementation against fine particulate air pollution in china. *J. Am. Coll. Cardiol.* 73, 2076–2085. [PubMed: 31023432]
- Macfarlane L-A, Murphy PR, 2010. MicroRNA: biogenesis, function and role in cancer. *Curr. Genom.* 11, 537–561.
- MacKinnon DP, Krull JL, Lockwood CM, 2000. Equivalence of the mediation, confounding and suppression effect. *Prev. Sci.* 1, 173–181. [PubMed: 11523746]
- MacKinnon DP, Lockwood CM, Hoffman JM, West SG, Sheets V, 2002. A comparison of methods to test mediation and other intervening variable effects. *Psychol. Methods* 7, 83–104. [PubMed: 11928892]
- Moore KJ, Rayner KJ, Suarez Y, Fernandez-Hernando C, 2011. The role of micrnas in cholesterol efflux and hepatic lipid metabolism. *Annu. Rev. Nutr.* 31, 49–63. [PubMed: 21548778]
- Motta V, Favero C, Dioni L, Iodice S, Battaglia C, Angelici L, et al. , 2016. Micrnas are associated with blood-pressure effects of exposure to particulate matter: results from a mediated moderation analysis. *Environ. Res* 146, 274–281. [PubMed: 26775008]
- Newby DE, Mannucci PM, Tell GS, Baccarelli AA, Brook RD, Donaldson K, et al. , 2015. Expert position paper on air pollution and cardiovascular disease. *Eur. Heart J.* 36, 83–93b. [PubMed: 25492627]
- Oh JG, Watanabe S, Lee A, Gorski PA, Lee P, Jeong D, et al. , 2018. Mir-146a suppresses sumo1 expression and induces cardiac dysfunction in maladaptive hypertrophy. *Circ. Res.* 123, 673–685. [PubMed: 30355233]
- Pergoli L, Cantone L, Favero C, Angelici L, Iodice S, Pinatel E, et al. , 2017. Extracellular vesicle-packaged mirna release after short-term exposure to particulate matter is associated with increased coagulation. *Part. Fibre Toxicol.* 14, 32. [PubMed: 28899404]
- Peters LJF, Biessen EAL, Hohl M, Weber C, van der Vorst EPC, Santovito D, 2020. Small things matter: Relevance of micrnas in cardiovascular disease. *Front. Physiol.* 11, 793. [PubMed: 32733281]
- Qin X, Guo J, 2020. MicroRNA-328–3p protects vascular endothelial cells against oxidized low-density lipoprotein induced injury via targeting forkhead box protein o4 (foxo4) in atherosclerosis. *Med. Sci. Monit.* 26, e921877. [PubMed: 32329448]
- Rodosthenous RS, Coull BA, Lu Q, Vokonas PS, Schwartz JD, Baccarelli AA, 2016. Ambient particulate matter and micrnas in extracellular vesicles: a pilot study of older individuals. *Part Fibre Toxicol.* 13, 13. [PubMed: 26956024]
- Rodosthenous RS, Kloog I, Colicino E, Zhong J, Herrera LA, Vokonas P, et al. , 2018. Extracellular vesicle-enriched micrnas interact in the association between long-term particulate matter and blood pressure in elderly men. *Environ. Res.* 167, 640–649. [PubMed: 30216846]
- Rotllan N, Price N, Pati P, Goedeke L, Fernandez-Hernando C, 2016. Micrnas in lipoprotein metabolism and cardiometabolic disorders. *Atherosclerosis* 246, 352–360. [PubMed: 26828754]

- Ruiz-Vera T, Ochoa-Martinez AC, Pruneda-Alvarez LG, Dominguez-Cortinas G, Perez-Maldonado IN, 2019. Expression levels of circulating micromas-126, -155, and -145 in mexican women exposed to polycyclic aromatic hydrocarbons through biomass fuel use. *Environ. Mol. Mutagen.* 60, 546–558. [PubMed: 30698845]
- Shen W, Weaver AM, Salazar C, Samet JM, Diaz-Sanchez D, Tong H, 2019. Validation of a dietary questionnaire to screen omega-3 fatty acids levels in healthy adults. *Nutrients* 11.
- Sobel ME, 1986. Some new results on indirect effects and their standard errors in covariance structure models. *Sociol. Methodol.* 16, 159–186.
- Sproston NR, Ashworth JJ, 2018. Role of c-reactive protein at sites of inflammation and infection. *Front Immunol.* 9, 754. [PubMed: 29706967]
- Sun D, Zhang J, Xie J, Wei W, Chen M, Zhao X, 2012. Mir-26 controls lxr-dependent cholesterol efflux by targeting abca1 and arl7. *FEBS Lett.* 586, 1472–1479. [PubMed: 22673513]
- Tong H, Rappold AG, Diaz-Sanchez D, Steck SE, Berntsen J, Cascio WE, et al. , 2012. Omega-3 fatty acid supplementation appears to attenuate particulate air pollution-induced cardiac effects and lipid changes in healthy middle-aged adults. *Environ. Health Perspect.* 120, 952–957. [PubMed: 22514211]
- Tong H, Zhang S, Shen W, Chen H, Salazar C, Schneider A, et al. , 2021. Lung function and short-term ambient air pollution exposure: Differential impacts of omega-3 and omega-6 fatty acids. *Ann. Am. Thorac. Soc.*
- WHO. 2020. The top 10 causes of death. Available: (<https://www.who.int/news-room/fact-sheets/detail/the-top-10-causes-of-death>) [accessed January 25th 2021].
- Xie Q, Wei M, Kang X, Liu D, Quan Y, Pan X, et al. , 2015. Reciprocal inhibition between mir-26a and nf-kappab regulates obesity-related chronic inflammation in chondrocytes. *Biosci. Rep.*
- Xing H, Peng M, Li Z, Chen J, Zhang H, Teng X, 2019. Ammonia inhalation-mediated mir-202–5p leads to cardiac autophagy through pten/akt/mtor pathway. *Chemosphere* 235, 858–866. [PubMed: 31284134]
- Zhao B, Johnston FH, Salimi F, Kurabayashi M, Negishi K, 2020. Short-term exposure to ambient fine particulate matter and out-of-hospital cardiac arrest: a nationwide case-crossover study in japan. *Lancet Planet. Health* 4, e15–e23. [PubMed: 31999950]
- Zhao C, Sun X, Li L, 2019. Biogenesis and function of extracellular mirnas. *ExRNA* 1, 38.

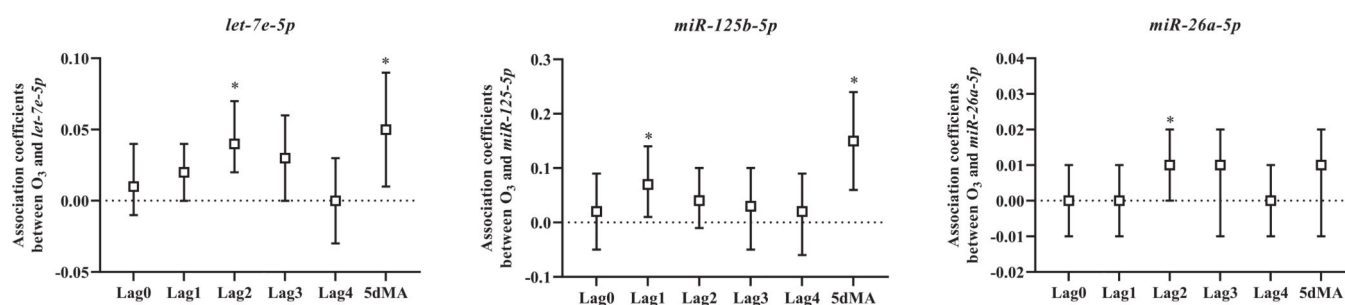
**Fig. 1.**

Flow of the statistical analysis and a schematic of the mediation model. A 4-step statistical analysis was shown in (A). In the mediation model (B), letters “a” indicates coefficients derived from the statistical analysis for “X → M”, “b” indicates coefficients for “M → Y”, and “c” indicates coefficients for X in “X + M → Y”.

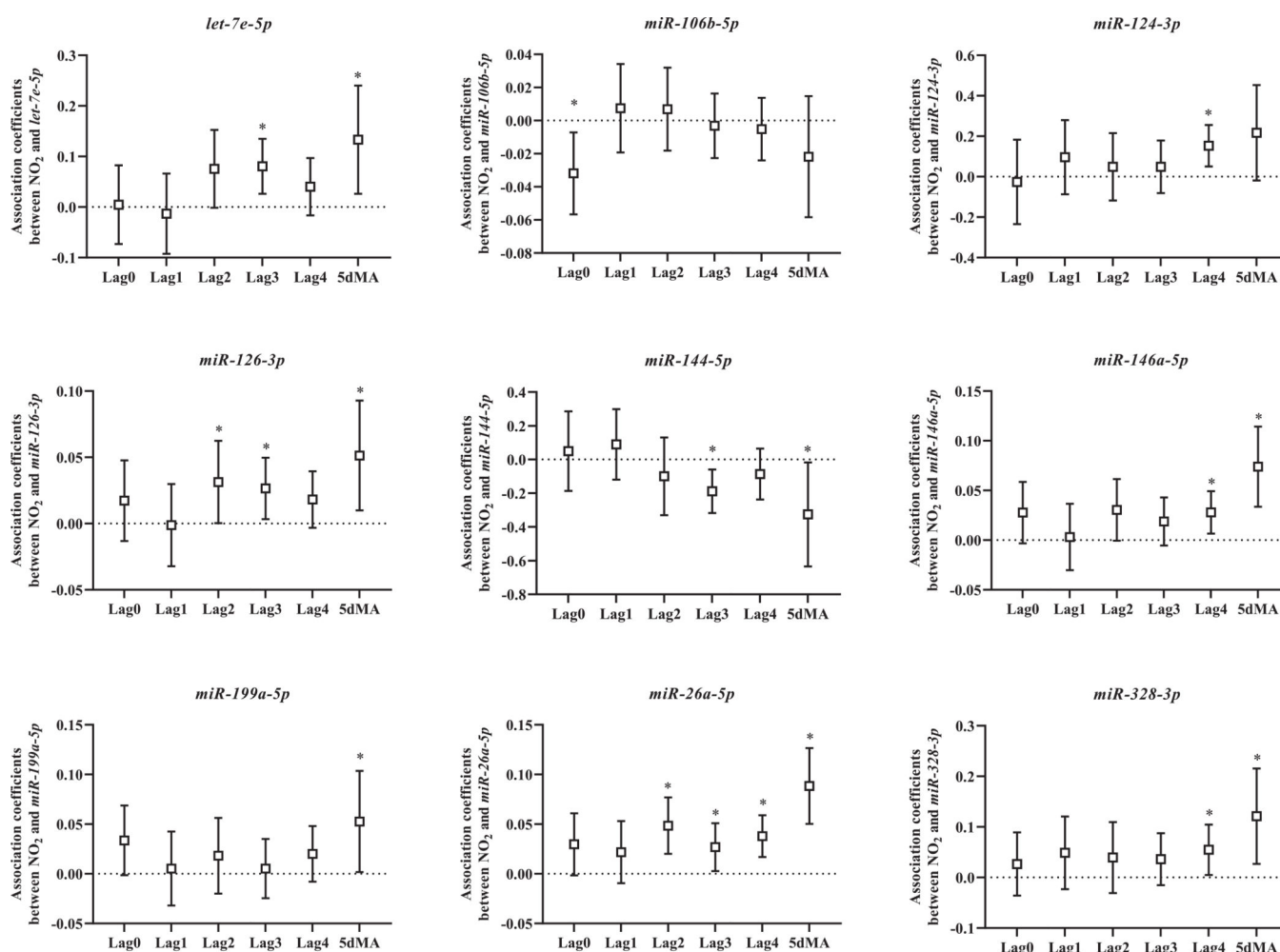
**Fig. 2.**

The association between short-term exposure to ambient PM_{2.5} and circulating miRNAs.

Coefficients (95% CI) between PM_{2.5} concentrations at lag0 - lag4 as well as 5dMA (5-day moving average) and changes in circulating *miR-124-3p*, *miR-125b-5p*, *miR-194-5p*, *miR-26a-5p*, *miR-320b*, *miR-34a-5p*, and *miR-370-3p* were derived from a generalized linear mixed model with random subject effects adjusted for omega-3 index age, sex, race, body mass index (BMI), long-term trend, day of week, relative humidity, and temperature. * indicates statistically significant association ($P < 0.05$). CI: confidence interval, PM_{2.5}: fine particulate matter.

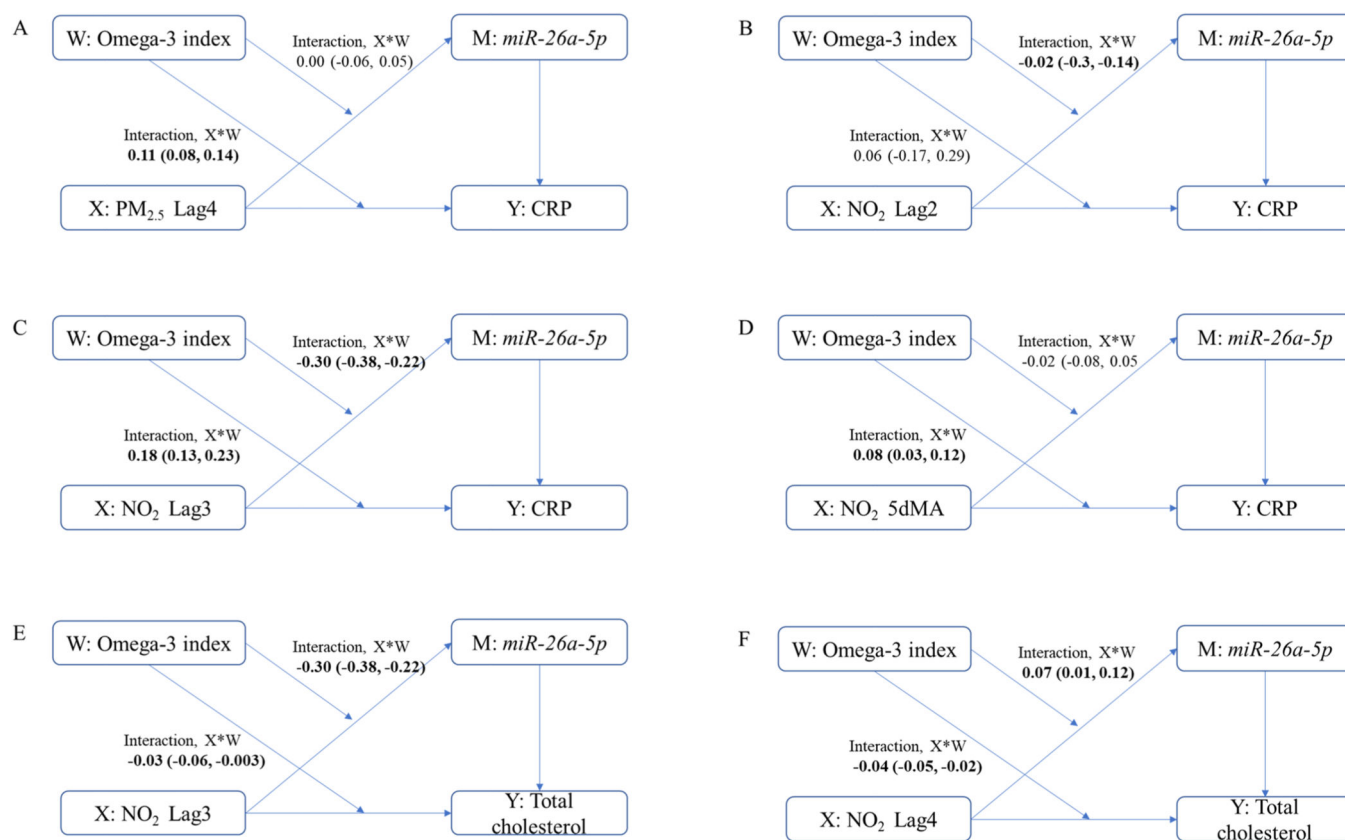
**Fig. 3.**

The association between short-term exposure to ambient O_3 and circulating miRNAs. Coefficients (95% CI) between O_3 concentrations at lag0 – lag4 as well as 5dMA (5-day moving average) and changes in circulating *let-7e-5p*, *miR-125b-5p*, and *miR-26a-5p* were derived from a generalized linear mixed model with random subject effects adjusted for omega-3 index age, sex, race, body mass index (BMI), long-term trend, day of week, relative humidity, and temperature. * indicates statistically significant association ($p < 0.05$). CI: confidence interval, O_3 : ozone.

**Fig. 4.**

The association between short-term exposure to ambient NO₂ and circulating miRNAs.

Coefficients (95% CI) between NO₂ concentrations at lag0 – lag4 as well as 5dMA (5-day moving average) and changes in circulating *let-7e-5p*, *miR-106b-5p*, *miR-124-3p*, *miR-126-3p*, *miR-144-5p*, *miR-146a-5p*, *miR-199a-5p*, *miR-26a-5p*, and *miR-328-3p* were derived from a generalized linear mixed model with random subject effects adjusted for omega-3 index age, sex, race, body mass index (BMI), long-term trend, day of week, relative humidity, and temperature. * indicates statistically significant association (P < 0.05). CI: confidence interval, NO₂: nitrogen dioxide.

**Fig. 5.**

The modifying effects of omega-3 index on the associations between air pollutants and miRNAs or between air pollutants and cardiovascular biomarkers. An interaction term between air pollutant and omega-3 index was introduced to the identified significant mediation pathways between air pollutants and downstream protein and lipid markers, including (A) PM_{2.5} Lag4 – *miR-26a-5p* – CRP, (B) NO₂ Lag2 – *miR-26a-5p* – CRP, (C) NO₂ Lag3 – *miR-26a-5p* – CRP, (D) NO₂ 5dMA – *miR-26a-5p* – CRP, (E) NO₂ Lag3 – *miR-26a-5p* – total cholesterol, (F) NO₂ Lag4 – *miR-26a-5p* – total cholesterol. Data were presented as coefficients with 95% confidence intervals and bold fonts indicate statistical significance ($P < 0.05$). 5dMA: 5-day moving average, CRP: c-reactive protein, NO₂: nitrogen dioxide, PM_{2.5}: fine particulate matter.

Table 1

Predicted targets of circulating miRNAs that were significantly associated with short-term exposure to ambient air pollution.

Air pollutant	Significantly associated miRNAs	Predicted targets
PM _{2.5}	<i>miR-124-3p</i>	IL-8, TNF α
	<i>miR-125b-5p</i>	SAA
	<i>miR-194-5p</i>	ET-1
	<i>miR-26a-5p</i>	CRP, Total Cholesterol, HDL
	<i>miR-320b</i>	D-dimer
	<i>miR-34a-5p</i>	CRP, sVCAM1
	<i>miR-370-3p</i>	CRP, Total Cholesterol, HDL
O ₃	<i>let-7e-5p</i>	CRP, IL-8, ET-1, IL-6
	<i>miR-125b-5p</i>	SAA
	<i>miR-26a-5p</i>	CRP, Total Cholesterol, HDL
NO ₂	<i>let-7e-5p</i>	CRP, IL-8, ET-1, IL-6
	<i>miR-106b-5p</i>	IL-8, TNF α
	<i>miR-124-3p</i>	IL-8, TNF α
	<i>miR-126-3p</i>	sVCAM1
	<i>miR-144-5p</i>	Total Cholesterol, HDL
	<i>miR-146a-5p</i>	CRP, IL-8
	<i>miR-199a-5p</i>	CRP
	<i>miR-26a-5p</i>	CRP, Total Cholesterol, HDL
	<i>miR-328-3p</i>	IL- β , IL-8, Ox-LDL

Table 2
Mediation effects of miRNAs on the association between exposure to air pollutants and biomarkers.

Air pollutant (X)	miRNAs (M)	Biomarker (Y)	X → M (a)	M → Y (b)	Direct effects (c')	Indirect effects (a*b)		
						Coefficient (95%CI)	Proportion (%)	p
Possible mediation								
PM _{2.5} lag4	miR-194-5p	ET-1	0.32 (0.01, 0.63)	-0.2 (-0.3, -0.11)	-0.1 (-0.19, 0)	-0.06 (-0.13, 0.00)	39.90	0.07
PM _{2.5} lag4	miR-26a-5p	CRP	0.46 (0.18, 0.74)	0.12 (0.08, 0.16)	0.11 (0.07, 0.15)	0.06 (0.02, 0.10)	34.14	0.005
PM _{2.5} 5dMA	miR-26a-5p	CRP	0.24 (-0.04, 0.53)	0.13 (0.1, 0.17)	0.08 (0.04, 0.12)	0.03 (-0.01, 0.07)	28.55	0.099
PM _{2.5} lag1	miR-320b	D-dimer	-0.17 (-0.21, -0.13)	-0.05 (-0.09, 0)	0.1 (0.07, 0.13)	0.01 (0.00, 0.02)	9.09	0.06
O ₃ lag2	miR-26a-5p	CRP	0.41 (0.01, 0.81)	0.09 (0.05, 0.12)	0.03 (-0.01, 0.08)	0.04 (0.00, 0.07)	50.81	0.06
O ₃ lag2	miR-26a-5p	Total Cholesterol	0.41 (0.01, 0.81)	-0.07 (-0.09, -0.05)	-0.16 (-0.19, -0.13)	-0.03 (-0.06, 0.00)	15.70	0.051
O ₃ lag2	miR-26a-5p	HDL	0.41 (0.01, 0.81)	-0.06 (-0.09, -0.02)	-0.13 (-0.18, -0.08)	-0.02 (-0.05, 0.00)	15.49	0.08
NO ₂ 5dMA	miR-146a-5p	CRP	0.64 (0.29, 0.98)	0.18 (0.01, 0.34)	0.02 (-0.21, 0.26)	0.11 (-0.01, 0.24)	82.37	0.07
NO ₂ 5dMA	miR-199a-5p	CRP	0.36 (0.01, 0.71)	0.07 (0.03, 0.1)	0.15 (0.1, 0.2)	0.02 (0.00, 0.05)	13.39	0.08
NO ₂ lag2	miR-26a-5p	CRP	0.52 (0.21, 0.82)	0.12 (0.09, 0.16)	0.07 (0.02, 0.13)	0.06 (0.02, 0.11)	46.81	0.003
NO ₂ lag3	miR-26a-5p	CRP	0.34 (0.04, 0.65)	0.14 (0.1, 0.18)	0.03 (-0.02, 0.08)	0.05 (0.00, 0.09)	61.22	0.04
NO ₂ lag3	miR-26a-5p	Total Cholesterol	0.34 (0.04, 0.65)	-0.1 (-0.12, -0.08)	-0.02 (-0.04, 0.01)	-0.04 (-0.07, -0.003)	68.60	0.03
NO ₂ lag3	miR-26a-5p	HDL	0.34 (0.04, 0.65)	-0.04 (-0.06, -0.02)	-0.08 (-0.1, -0.05)	-0.01 (-0.03, 0.00)	14.69	0.06
NO ₂ lag4	miR-26a-5p	Total Cholesterol	0.5 (0.22, 0.78)	-0.08 (-0.1, -0.05)	-0.04 (-0.07, -0.02)	-0.04 (-0.06, -0.01)	46.54	0.002
NO ₂ 5dMA	miR-26a-5p	CRP	0.7 (0.4, 1)	0.08 (0.03, 0.13)	0.13 (0.07, 0.18)	0.06 (0.02, 0.10)	30.76	0.01
Possible suppression								
PM _{2.5} lag4	miR-125b-5p	SAA	0.39 (0.06, 0.73)	-0.19 (-0.23, -0.15)	0.14 (0.1, 0.18)	-0.08 (-0.14, -0.01)	NA	0.03
PM _{2.5} 5dMA	miR-125b-5p	SAA	0.3 (0.02, 0.59)	-0.12 (-0.16, -0.08)	0.03 (-0.01, 0.07)	-0.04 (-0.07, 0.00)	NA	0.05
PM _{2.5} lag4	miR-34a-5p	CRP	0.41 (0.07, 0.75)	-0.05 (-0.08, -0.01)	0.22 (0.17, 0.26)	-0.02 (-0.04, 0.00)	NA	0.09
PM _{2.5} lag4	miR-34a-5p	sVCAM1	0.41 (0.07, 0.75)	-0.15 (-0.19, -0.1)	0.14 (0.09, 0.2)	-0.06 (-0.11, -0.01)	NA	0.02
O ₃ lag1	miR-125b-5p	SAA	0.55 (0.07, 1.02)	-0.07 (-0.1, -0.04)	0.02 (-0.03, 0.07)	-0.04 (-0.07, 0.00)	NA	0.04
NO ₂ lag3	miR-144-5p	HDL	-0.59 (-0.69, -0.49)	-0.03 (-0.04, -0.01)	-0.08 (-0.11, -0.06)	0.02 (0.01, 0.03)	NA	0.002
NO ₂ lag0	miR-199a-5p	CRP	0.33 (-0.01, 0.68)	0.07 (0.04, 0.09)	-0.01 (-0.04, 0.03)	0.02 (0.00, 0.05)	NA	0.08
NO ₂ lag4	miR-26a-5p	CRP	0.5 (0.22, 0.78)	0.14 (0.1, 0.18)	-0.01 (-0.06, 0.03)	0.07 (0.03, 0.11)	NA	0.002

Air pollutant (X)	miRNAs (M)	Biomarker (Y)	X → M (a)	M → Y (b)	Direct effects (c')	Indirect effects (a*b)	
						Coefficient (95% CI)	Proportion (%)
NO ₂ 5dMA	<i>miR-328-3p</i>	Ox-LDL	0.41 (0.09, 0.73)	0.08 (0.01, 0.15)	-0.2 (-0.29, -0.12)	0.03 (-0.01, 0.07)	NA
							0.09

The letters X, M, and Y denote independent variable (i.e., air pollutant), mediator (i.e., miRNAs), and dependent variable (i.e., protein / lipid biomarkers) in Fig. 1. The letters a, b, and c' denote coefficients of X, M, and X in their respective models. Data were presented as the coefficients with 95% confidence interval (95%CI). The statistical significance of indirect effects (a*b) was assessed using the Sobel Test. "Significant (p < 0.05)" with bold fonts and borderline significant (p < 0.10) results were presented in this table. Results with different directions of change between direct and indirect effects were considered as possible suppression. Proportion of indirect effects in the total effects (a*b+c') was calculated only for those indicating possible mediation. CRP: C-reactive protein, ET-1: endothelin 1, HDL: high-density lipoprotein, NO2: nitrogen dioxide, O3: ozone, Ox-LDL: oxidized low-density lipoprotein, PM2.5: fine particulate matter, SAA: serum amyloid A, sVCAM1: soluble vascular adhesion molecule 1, TNFα: tumor necrosis factor alpha.